

CS 540 Spring 2024 Midterm

Exam Version: 1

First, Last Name:

Section:

Net ID: _____@wisc.edu

Campus ID Number:

1. You cannot sit next (front, back, left (within two seats including an empty seat), right (within two seats including an empty seat)) to someone you know. Switch seats now if that's the case.
2. Fill in these fields (left to right) on the scantron sheet using a pencil:
 - LAST NAME (family name) and FIRST NAME (given name), fill in bubbles
 - IDENTIFICATION NUMBER is your Campus ID number, fill in bubbles
 - Under A, B, C of SPECIAL CODES, fill in your three-digit section number (001, 002, 003)
 - **Under F of SPECIAL CODES, fill in Exam version above.** This is very important!

Make sure you fill all the above accurately in order to get graded.

3. Mark your answers on the Scantron sheet **and this handout**. Use a #2 pencil to mark all answers. Each question has exactly one correct answer.
4. You may only reference your one-sheet notes. Please turn off and put away portable electronics now.
5. If you need to use the restroom, your phone must remain in the room, placed face down on your desk and kept visible to the proctors.
6. You have 90 minutes to take the exam. Once you have finished, please show your student ID to one of the proctors and submit both your scantron and this handout.

Good luck!

1. You are building a spam filter for an email system. Your filter does this by using the word "sale" in the email as evidence. What is the probability of the email being spam knowing the following: (Round to the nearest percent).
- The prior probability of a spam email is 4%
 - The probability of the word "sale" in a spam email is 85%
 - The probability of the word "sale" in a non-spam email is 15%
- A. 13%
- B. 15%
- C. 17%
- D. 19%
- E. None of the above.

Solution: D -

$$19\% \approx \frac{0.85 * 0.04}{0.85 * 0.04 + 0.15 * 0.96} * 100$$

ALL ANSWERS WERE ACCEPTED(FREE POINT): There was confusion on what the question was asking you to calculate.

2. You want to build a text generator with a language model using the n-gram method. What is NOT a difference between a 10-gram model and a 5-gram model?
- A. 10-gram models capture longer range dependencies between words.
- B. 10-gram models can generate more expressive sequences.
- C. There are fewer 10-gram sequences in a 500-word body of text than 5-gram sequences.
- D. Smoothing is more useful for a 5-gram sequence
- E. All of the above are differences.

Solution: D - Smoothing is more useful as n increased so it is more useful for a 10-gram instead of a 5-gram

3. For which of the following tasks is PCA a viable choice?
- A. Reducing the memory storage requirement for a dataset.
- B. Visualization.
- C. Better understanding patterns and trends in a dataset.
- D. All of the above.

E. None of the above.

Solution: D - PCA is a method of dimensionality reduction, where each additional component “explains” as much of the data as possible. Projecting to these components reduces the size of the dataset and allows for visualization in 2- or 3-dimensional space, while the components themselves may be analyzed to better understand the salient features of the dataset.

4. When interpreting a dendrogram in hierarchical clustering, how do we determine which clusters are most similar?
- A. Compare the lengths of the branches leading to each cluster. Longer branches indicate higher similarity.
 - B. Look for clusters connected by the lowest branches on the dendrogram.
 - C. Identify clusters at the same horizontal level on the dendrogram.
 - D. Analyze the color or shading used to represent different clusters on the dendrogram (if applicable).
 - E. None of the above.

Solution: B - The dendrogram represents the hierarchical merging process. Clusters connected by the lowest branches merged at the earliest point in the process, meaning they had the smallest dissimilarity (highest similarity) at that point. These clusters were therefore the most similar before any further merging occurred.

5. In K-means clustering, how are data points assigned to clusters in each iteration?
- A. They are assigned to the cluster with agglomerative distance under complete linkage.
 - B. They are assigned to the cluster with the nearest centroid.
 - C. They are assigned randomly to each cluster.
 - D. They remain assigned to the same cluster as the previous iteration.
 - E. None of the above.

Solution: B - In each iteration of K-means clustering, data points are re-assigned to the cluster with the closest centroid. This is calculated by measuring the distance (e.g., Euclidean) between each data point and all centroids, and assigning the point to the cluster with the minimum distance. This process iterates until the centroids and cluster assignments no longer change significantly, or a specified number of iterations is reached.

6. Which step is **NOT** a part of the K-means clustering algorithm?
- A. Randomly initialize cluster centroids.
 - B. Assign each data point to the nearest centroid.
 - C. Recompute cluster centroids based on the mean of assigned data points.
 - D. Update the number of cluster centroids.
 - E. None of the above.

Solution: D - Updating the number of clusters is not included in the K-means algorithm. The algorithm operates with a pre-defined number of clusters (K) throughout the iterations. Determining the optimal number of clusters is a separate problem, and K-means itself doesn't automatically adjust this value.

7. You have a data set for regression such that:

- $x_1, y_1 = ([1, 2, 3, 4, -5], -1)$
- $x_2, y_2 = ([2, 6, 4, 8, 10], 1)$
- $x_3, y_3 = ([1, 3, 5, 7, 9], -1)$

What are the labels, number of points (n), and dimension of the features (d)?

- A. The labels are -1 and 1 ; $n = 3$; $d = 5$.
- B. The labels are odd and even; $n = 5$; $d = 4$.
- C. The labels are $[1, 2, 3, 4, -5], [2, 6, 4, 8, 10], [1, 3, 5, 7, 9]$; $n = 3$; $d = 4$.
- D. The labels are -1 and 1 ; $n = 4$; $d = 3$.
- E. None of the above.

Solution: A. There are 3 data points and each datapoint has 5-dimensional features. The labels are the -1 and 1 values.

8. Consider the linear regression model: $\hat{y} = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3$. Using a set of weights $\theta_0 = 3, \theta_1 = 1, \theta_2 = 2, \theta_3 = 1$, predict $\hat{y}$ for the test point $x = [1, 2, 3]$. Also, using the weights, compute the mean squared error (MSE) on the training set:

- $x_1, y_1 = ([-1, 0, 1], 2)$
- $x_2, y_2 = ([1, 2, 2], 13)$
- A. $\hat{y} = 16, \text{MSE} = 5$
- B. $\hat{y} = 11, \text{MSE} = 5$
- C. $\hat{y} = 16, \text{MSE} = 10$

- D. $\hat{y} = 11$, $\text{MSE} = 10$
 E. None of the above.

Solution: B. $\hat{y} = 1 \times 1 + 2 \times 2 + 3 \times 1 + 3 = 11$ and $\text{MSE} = \frac{(3-2)^2 + (13-10)^2}{2} = 5$.

9. How can we improve the computational performance of Naive Bayes?
- I Compute in the log domain by multiplying the log-probabilities to avoid underflow
 - II Compute in the log domain by adding the log-probabilities to avoid underflow
 - III Ignore terms that do not depend on the class and normalize at the end
- A. I Only
 B. II Only
 C. I and III
 D. II and III
 E. None of the above.

Solution: D - Computing in the log domain improves computational importance by avoiding underflow. Recall the log rules where the log of the product of the probabilities becomes the the sum of the logs of the probabilities. I incorrectly states that we should multiply these log-probabilities while II correctly states that we should add the log-probabilities. Additionally, we can avoid computing unnecessary terms that are independent of the class. Instead of computing these terms to directly find probabilities, we can instead find a term proportional to the true posterior probability and normalize among them to recover the probabilities., so III is correct.

10. Consider a dataset with two features: X_1 which can have two possible values a_1 or a_2 and X_2 which can have two possible values b_1 or b_2 . There are two possible classifications for the datapoints: $Y = y_1$ and $Y = y_2$. Given the following frequency table and **using Naive Bayes**, what should we label a datapoint with $X_1 = a_2$ and $X_2 = b_2$, and with what posterior probability? *Hint: the frequency table contains the number of samples we have access to for each combination of features and labels.*

	$X_1 = a_1, X_2 = b_1$	$X_1 = a_1, X_2 = b_2$	$X_1 = a_2, X_2 = b_1$	$X_1 = a_2, X_2 = b_2$
$Y = y_1$	0	200	200	0
$Y = y_2$	90	10	60	40

- A. Label y_1 , probability 0.8
 B. Label y_2 , probability 0.8
 C. Label y_2 , probability 1

- D. Label y_1 , probability 0.6
- E. None of the above.

Solution: A - First, figure out the posterior probabilities. $P(Y = y_1|X_1 = a_2, X_2 = b_2) \propto P(X_1 = a_2|Y = y_1)P(X_2 = b_2|Y = y_1)P(Y = y_1) = 200/400 * 200/400 * 400/600 = 1/6$ and $P(Y = y_2|X_1 = a_2, X_2 = b_2) \propto P(X_1 = a_2|Y = y_2)P(X_2 = b_2|Y = y_2)P(Y = y_2) = 100/200 * 50/200 * 200/600 = 1/24$. Then, normalize these to get $P(Y = y_1|X_1 = a_2, X_2 = b_2) = \frac{1/6}{1/6+1/24} = 4/5$. We can see that $P(Y = y_2|X_1 = a_2, X_2 = b_2) = 1/5$, so we would label this point y_1 with posterior probability $4/5 = 0.8$.

11. Consider the following probabilities: $P(B|A) = \frac{1}{6}$, $P(A|B) = \frac{1}{4}$, and $P(B) = \frac{1}{2}$. What is $P(A)$?
- A. $\frac{1}{3}$
 - B. $\frac{3}{4}$
 - C. $\frac{2}{3}$
 - D. $\frac{1}{2}$
 - E. None of the above.

Solution: B - Use Bayes' Rule. $P(B|A) = \frac{P(A|B)P(B)}{P(A)}$, so $P(A) = \frac{P(A|B)P(B)}{P(B|A)}$ Substitute values in to get $P(A) = \frac{1/4 * 1/2}{1/6} = \frac{1/8}{1/6} = \frac{3}{4}$

12. Which of the following statements about Naive Bayes is **true**?
- A. Multiplying many probabilities runs the risk of underflow.
 - B. All features of the dataset are equally important.
 - C. Basic Naive Bayes cannot predict a label not seen in its dataset.
 - D. Naive Bayes predicts the label that maximizes the posterior distribution.
 - E. All of the above.

Solution: E - The first statement is true: when a lot of probabilities are multiplied together, the product approaches 0 and can be so small the computer can no longer accurately represent it. The second statement is also true because there are no weightings on feature probabilities in the likelihood. The third statement is true. While there are ways to mitigate this, if a label is not seen in the dataset, all likelihood and prior probabilities are 0 and the posterior is 0. The fourth statement is true by definition.

13. Consider the perceptron $f(x) = \sigma(w^T x + b)$, where $w = [-1, 4]$, $b = -5$, and the activation function is ReLU: $\sigma(z) = \max(0, z)$. Compute its output on the following two data points: $x_1 = [-2, 3]$, $x_2 = [15, 5]$.
- A. $f(x_1) = 9, f(x_2) = 0$
 - B. $f(x_1) = -9, f(x_2) = 1$
 - C. $f(x_1) = 0, f(x_2) = 1$
 - D. $f(x_1) = 9, f(x_2) = 10$
 - E. None of the above.

Solution: A $f(x_1) = \max(0, -1 \times -2 + 4 \times 3 - 5) = 9, f(x_2) = \max(0, -1 \times 15 + 4 \times 5 - 5) = 0$

14. Suppose you are given a 4-layer multilayer perceptron (3 hidden layers h_1, h_2 , and h_3 and 1 output layer). **All layers involve bias terms.** All activation functions are sigmoids, and the output layer uses a softmax function. Suppose h_1 has 300 units, h_2 has 150 units, and h_3 has 100 units. Given a dataset with 10 input features and 5 unique class labels, how many learnable parameters does the perceptron have in total?
- A. 63,500
 - B. 64,055
 - C. 63,550
 - D. 64,050
 - E. None of the above.

Solution: B. $10 \times 300 + 300 + 300 \times 150 + 150 + 150 \times 100 + 100 + 100 \times 5 + 5 = 64,055$

15. Given Boolean (0, 1 encoding) variables x_1, x_2, x_3 , define ternary operator $x_1 ? x_2 : x_3$ to output the Boolean value “if x_1 then x_2 else x_3 ”. What weights in a single perceptron with input $x = (x_1, x_2, x_3)^T$ and step function activation implement this ternary operator?
- A. Cannot be implemented by such a perceptron
 - B. $w_1 = 1, w_2 = 1, w_3 = 1, b = 2.5$
 - C. $w_1 = 1, w_2 = 1, w_3 = 1, b = 1.5$
 - D. $w_1 = 1, w_2 = 1, w_3 = 1, b = 0.5$
 - E. None of the above.

Solution: A. This is a slight generalization to the XOR example we learned in class. Place the input in a 3D space: x_1 on the x-axis and can take value either 0 or 1; x_2 on the y-axis, and x_3 on the z-axis. Then, at the coordinate specified by (x_1, x_2, x_3) place the operator's value. For example, at $(1, 0, 1)$ the operator $x_1 ? x_2 : x_3$ evaluates to x_2 's value which is 0. Now inspect the resulting cube. You will notice there is a 2D slice that contains the XOR problem. A single perceptron produces a linear decision boundary plane in 3D; within the 2D slice the decision boundary is a line. Therefore, the perceptron cannot solve the problem within the 2D slice. Consequently, it cannot solve the overall problem.

16. Which of the following would you consider to be valid activation functions (**elementwise non-linearities**) to train a neural net in practice?

- A. $f(x) = -x - 15$
- B. $f(x) = 10x + 10$
- C. $f(x) = \begin{cases} \min(x, 0.1x), & x \geq 0 \\ \min(x, 0.1x), & x < 0 \end{cases}$
- D. $f(x) = -\min(2, x)$
- E. None of the above.

Solution: C or D. The first two are linear functions.
If you choose either option C or D, you will receive full credit.

17. In a 3-layer neural network, suppose that the hidden layers use the Sigmoid activation function, while the output layer uses the Rectified Linear Unit (ReLU) activation function. Which statement is true?

- A. The neural network output is always between -1 and 1 regardless of input.
- B. The neural network output is always non-zero regardless of input.
- C. The neural network output is always non-negative regardless of input.
- D. The neural network output can be any real number regardless of input.
- E. None of the above.

Solution: C. The Sigmoid activation function in the hidden layers squashes the output values between 0 and 1, making them non-negative. The ReLU activation function in the output layer outputs the input directly if it is positive and zero otherwise. Since the input to the ReLU activation function is from the Sigmoid function, which outputs non-negative values, the output of the ReLU function will also be non-negative.

18. Given input x , a perceptron outputs $\sigma(w^T x + b)$. Let $w = (1, 2)^T$, $b = -1$, and σ is a leaky ReLU defined as $f(x) = \max(0.3x, x)$. What is the perceptron's output given input $x = (0, 0)^T$, $(0, 1)^T$, and $(2, 1)^T$, respectively?
- A. 0, 1, 3
 - B. -0.3, 0.3, 3
 - C. -1, 1, 3
 - D. -0.3, 1, 3
 - E. None of the above.

Solution: D.

For input $x = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$:

$$w^T x + b = (1 * 0 + 2 * 0) - 1 = -1$$

$$f(-1) = \max(0.3 * (-1), -1) = \max(-0.3, -1) = -0.3$$

For input $x = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$:

$$w^T x + b = (1 * 0 + 2 * 1) - 1 = 1$$

$$f(1) = \max(0.3 * 1, 1) = \max(0.3, 1) = 1$$

For input $x = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$:

$$w^T x + b = (1 * 2 + 2 * 1) - 1 = 3$$

$$f(3) = \max(0.3 * 3, 3) = \max(0.9, 3) = 3$$

19. Which of the following statement(s) about KNN is/are correct? **Circle all correct statements.**
- A. For KNN, the accuracy of the algorithm increases for larger k .
 - B. One advantage of KNN is that the algorithm runs fast for even very large datasets.
 - C. The KNN algorithm can be used for both supervised learning and unsupervised learning.
 - D. The output of KNN is invariant under any linear transformation for the sample space.
 - E. None of the above.

Solution: E. The accuracy of KNN does not necessarily increase for larger K . KNN runs slow for very large dataset since it needs to store the data. KNN is a supervised learning algorithm. Some linear transformation will change the distance between the points therefore change the output of KNN.

20. We are given two training points in 3D: $x_1 = (0, 0, 0)$, $y_1 = -$ and $x_2 = (0, 2, 2)$, $y_2 = +$. Suppose we are running 1NN for the input $x = (0, 0, 3)$ using Euclidean distance, what is the distance between x and the nearest neighbor in the dataset and what is going to be the output of the algorithm?
- A. 1, $-$.
 - B. $\sqrt{3}$, $+$.
 - C. 3, $-$.
 - D. $\sqrt{5}$, $+$.
 - E. None of the above.

Solution: D.

For $x_1 = (0, 0, 0)$:

$$\text{Distance} = \sqrt{(0-0)^2 + (0-0)^2 + (3-0)^2} = \sqrt{0+0+9} = \sqrt{9} = 3$$

For $x_2 = (0, 2, 2)$:

$$\text{Distance} = \sqrt{(0-0)^2 + (0-2)^2 + (3-2)^2} = \sqrt{0+4+1} = \sqrt{5}$$

The nearest neighbor is x_2 , and the distance between x and x_2 is $\sqrt{5}$.

The output of the 1-nearest neighbor (1NN) algorithm is determined by the label of the nearest neighbor. Since x_2 is the nearest neighbor and its label is $+$, the output of the algorithm will be $+$.